

**UNIVERSIDAD POLITÉCNICA DE
SAN LUIS POTOSÍ**

CARRERA: INGENIERIA EN TECNOLOGIAS DE LA INFORMACION

MATERIA: DESARROLLO DEL PENSAMIENTO CRITICO

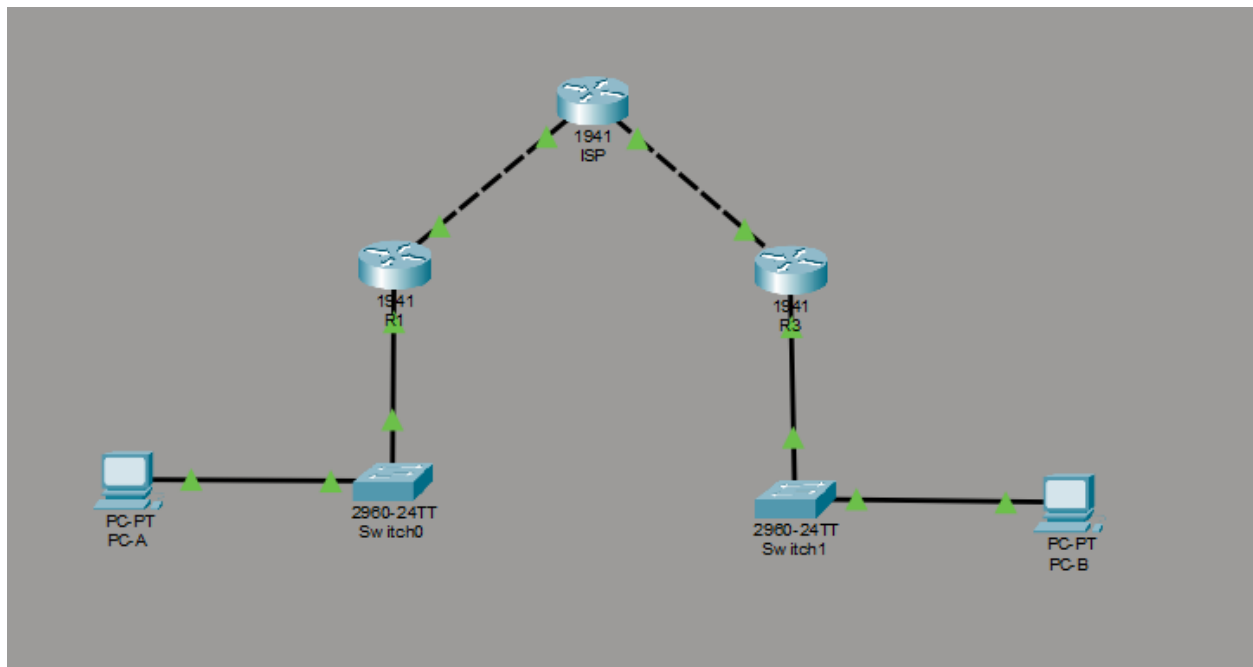
TRABAJO: Act 6

PARCIAL: 1

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Profesor: Servando Lopexz Contreras

Topología de red



Configuración inicial

configuración del router 1

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco IOS1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface G0/0
R1(config-if)# ip address 209.165.100.1 255.255.255.0
R1(config-if)# no shutdown

R1(config-if)#exit
R1(config)#interface G0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown

R1(config-if)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Configuración del router 3

```
R3

Physical Config CLI Attributes

IOS Command Line Interface

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

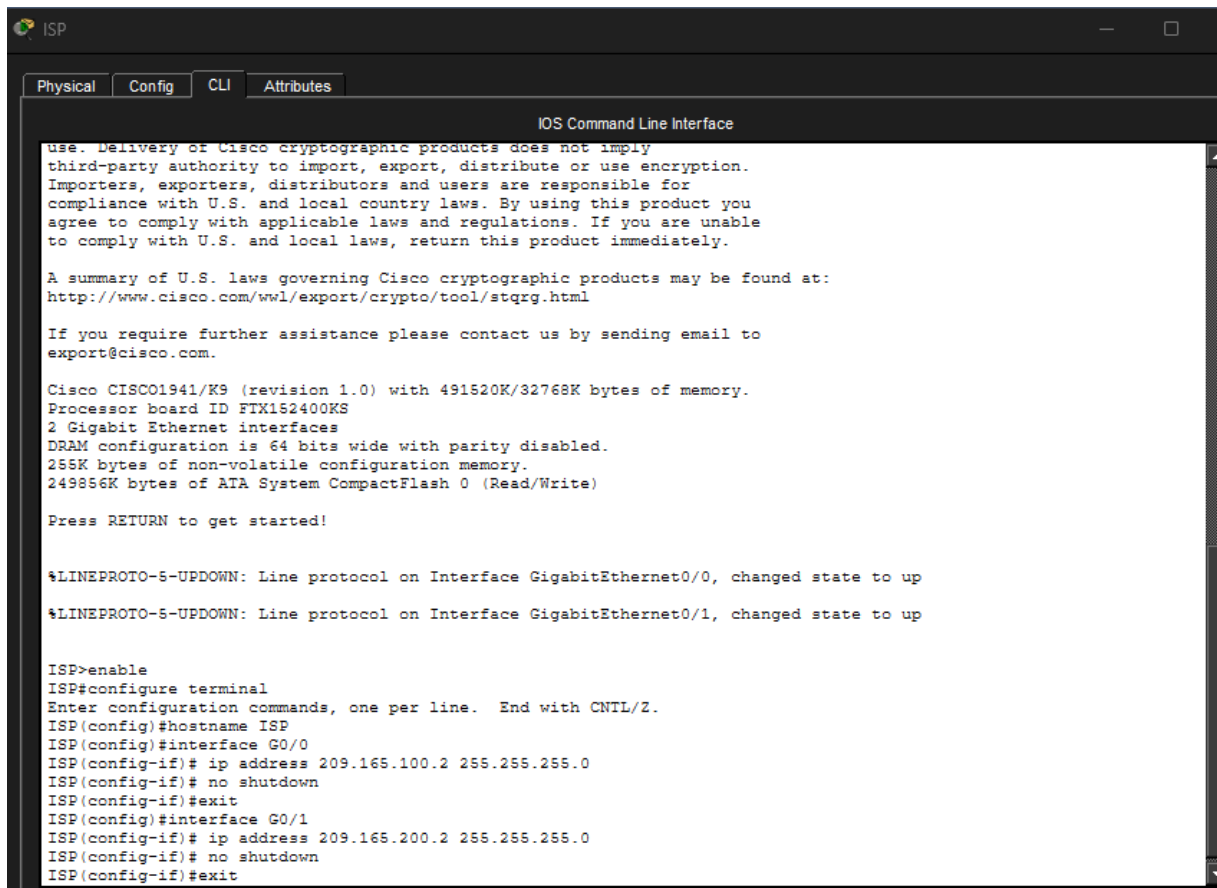
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface G0/0
R3(config-if)# ip address 209.165.200.1 255.255.255.0
R3(config-if)# no shutdown

R3(config-if)#exit
R3(config)#interface G0/1
R3(config-if)# ip address 192.168.3.1 255.255.255.0
R3(config-if)# no shutdown

R3(config-if)#exit
R3(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.2
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Configuracion de ISP



The image shows a terminal window titled "ISP" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The text in the terminal includes a disclaimer about Cisco cryptographic products, a link to U.S. laws, contact information, and system details for a Cisco CISC01941/K9. It also shows the configuration of two Gigabit Ethernet interfaces (G0/0 and G0/1) with IP addresses 209.165.100.2 and 209.165.200.2 respectively, both with a subnet mask of 255.255.255.0 and no shutdown.

```
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

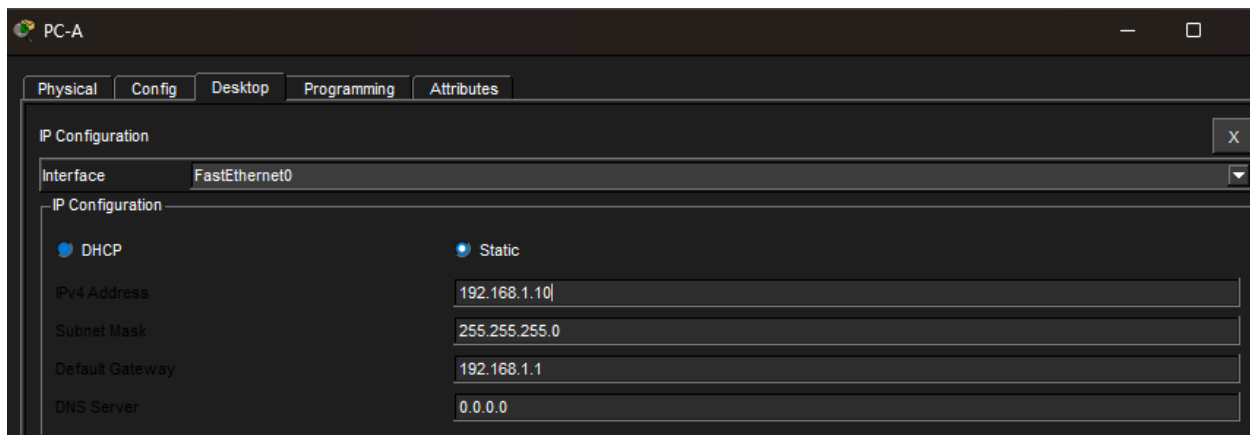
Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

ISP>enable
ISP#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP(config)#hostname ISP
ISP(config)#interface G0/0
ISP(config-if)# ip address 209.165.100.2 255.255.255.0
ISP(config-if)# no shutdown
ISP(config-if)#exit
ISP(config)#interface G0/1
ISP(config-if)# ip address 209.165.200.2 255.255.255.0
ISP(config-if)# no shutdown
ISP(config-if)#exit
```

PC-A configuración



The image shows a window titled "PC-A" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Config" tab is active, displaying the "IP Configuration" window. The "Interface" is set to "FastEthernet0". The "IP Configuration" section shows "DHCP" and "Static" radio buttons, with "Static" selected. The "IPv4 Address" is 192.168.1.10, the "Subnet Mask" is 255.255.255.0, the "Default Gateway" is 192.168.1.1, and the "DNS Server" is 0.0.0.0.

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

PC-B configuración



3 Implementación de ACLs

```
access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.2554
```

3 Phase 01: ISAKMP Policy

```
crypto isakmp policy 10
```

```
encryption aes 256
```

```
authentication pre-share
```

```
group 5
```

```
exit
```

```
crypto isakmp key secretkey address 209.165.100.1
```

5 Phase 02: IPSec Transform-Set

```
crypto ipsec transform-set R3-R1 esp-aes 256 esp-sha-hmac
```

6 Crear el Mapa Criptográfico

```
crypto map IPSEC-MAP 10 ipsec-isakmp
```

```
set peer 209.165.100.1
```

```
set transform-set R3-R1
```

```
set pfs group5
```

```
set security-association lifetime seconds 86400
```

```
match address 100
```

```
exit
```

7 Aplicar el Mapa Criptográfico

```
int g0/0
```

```
crypto map IPSEC-MAP
```

```
exit
```

Conclusión

Para asegurar la conexión, primero habilité la licencia de seguridad y configuré la ACL 100 para identificar el tráfico que debe protegerse. Luego, establecí las políticas de negociación en la Fase 1 y definí el método de cifrado de datos en la Fase 2 mediante el Transform-Set. Finalmente, integré todo en un Mapa Criptográfico que vincula al router destino con la lista de acceso, aplicándolo directamente en la interfaz de salida.